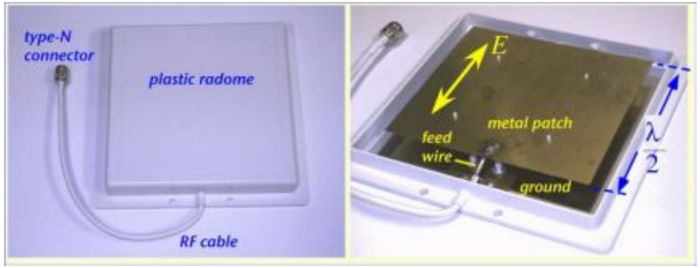
Antenna Fundamentals

Antennas


type-N
connector
plastic radome
RF cable
E
metal patch
feed
wire
ground
$\frac{\lambda}{2}$

Antenna Radiation Pattern

- Captures the relative field strength of the radio waves in a given directional
  - Horizontal pattern (X-Y, azimuth): parallel to the ground
  - Vertical pattern (Y-Z, elevation): perpendicular to the ground

- Reciprocity Property: Fundamental properties of an antenna are same whether it is transmitting or receiving
  - E.g. receiving pattern is identical to radiating pattern

Antenna Radiation Pattern

- Omnidirectional: radiates energy uniformly in all directions in a given plane
  - Dipole, whip, rubber duck

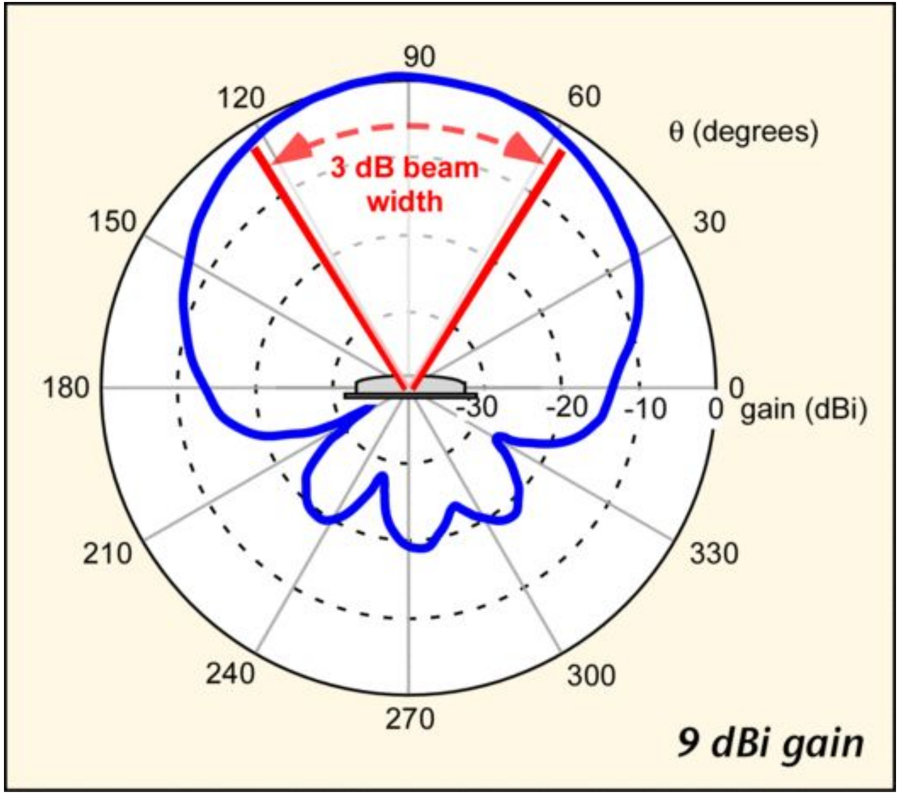
- Directional: radiates energy in some direction more than others
  - Yagi-uda, patch, sector, dish

Antenna Terminology

- Antenna Gain
- 3dB Beam width
- Front-to-back ratio

Antenna Gain

- Ratio of power gain in a given direction to the
- power gain of a reference antenna in the same
- direction.
- • Reference antenna: isotropic antenna (radiates
- power equally in all directions)
- – Isotropic antennas don’t exist
- – Real antennas have directionality



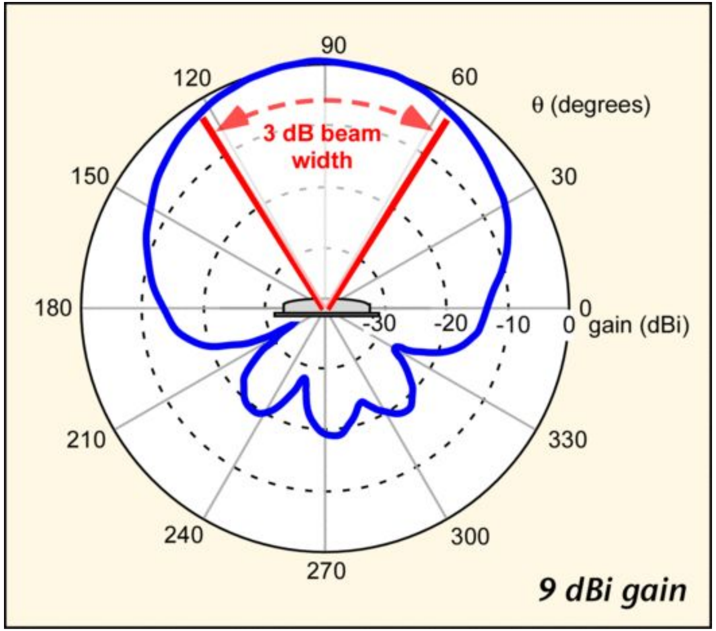
90
120
60
θ (degrees)
3 dB beam width
150
30
180
0
-30
-20
-10
0
gain (dBi)
210
330
240
300
270
9 dBi gain

Antenna Gain

- Antennas with gain don’t create radiated power.
  - They just distribute the power differently relative to radiating the power equally in all directions

3dB Beam Width

3dB beam width (half-power): Angle between the points in the main lobe that are down from the maximum gain by 3 dB



Front-to-back ratio

- Applicable to mostly directional antennas
  - Ratio of the peak gain in the forward direction to the gain 180-degrees behind the peak